

ATYRAU INTL, Kazakhstan

WMO: 381411

Lat: 47.122N

Lon: 51.821E

Elev: -72

StdP: 14.73

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-4.2	-14.9	2.4	-10.2	-9.0	3.4	-4.4	27.7	27.4	25.6	24.9	6.4	50	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	100.7	67.8	98.5	67.6	95.3	66.7	74.2	85.5	72.1	85.7	70.4	86.3	12.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	115.2	80.2	68.1	102.9	77.5	65.9	95.2	76.4	37.7	84.3	35.7	85.4	34.2	85.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
26.7	23.6	21.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-14.1	103.6	8.2	3.1	-20.0	105.8	-24.8	107.6	-29.4	109.3	-35.4	111.6
			-14.4	78.6	8.1	2.9	-20.2	80.7	-24.9	82.3	-29.4	83.9	-35.3	86.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	17.9	19.8	36.2	52.4	67.2	76.2	80.1	77.9	64.9	49.3	35.1	25.6		
	DBStd	23.72	11.63	12.08	7.66	8.91	7.92	7.28	6.07	7.44	7.67	8.36	9.35	9.33		
	HDD50	3668	995	846	432	74	1	0	0	0	2	114	449	755		
	HDD65	6783	1460	1266	892	389	66	7	0	5	94	488	896	1220		
	CDD50	3816	0	0	4	146	536	787	932	866	449	93	3	0		
	CDD65	1456	0	0	0	10	135	344	468	405	92	2	0	0		
	CDH74	19639	0	0	1	124	1763	4678	6509	5510	1034	20	0	0		
	CDH80	10362	0	0	0	29	746	2505	3665	3074	342	1	0	0		
(14) Wind	WSAvg	11.0	11.3	11.5	12.6	12.4	11.6	10.8	10.3	9.9	10.3	10.2	10.1	11.4		
(15) Precipitation	PrecAvg	7.2	0.6	0.5	0.6	0.6	1.1	0.7	0.4	0.4	0.4	0.6	0.7	0.7		
	PrecMax	13.4	1.3	1.1	1.9	1.8	3.5	1.8	1.6	1.5	1.6	1.9	2.0	1.9		
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	2.0	0.4	0.3	0.5	0.4	0.9	0.6	0.4	0.5	0.4	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	43.0	64.1	84.4	96.4	102.5	104.0	104.1	93.1	76.8	60.4	45.0		
		MCWB	38.3	38.6	50.2	60.0	63.0	66.0	68.3	67.2	64.3	57.5	53.7	42.1		
	2%	DB	36.0	39.0	57.3	77.1	91.7	98.9	100.6	100.7	87.6	71.8	55.4	42.6		
		MCWB	35.8	36.2	47.2	57.7	63.2	66.8	68.5	67.8	61.7	55.6	49.9	40.5		
	5%	DB	34.1	35.9	52.1	72.0	88.0	96.4	98.4	98.3	83.9	67.8	51.8	39.3		
		MCWB	33.9	34.4	44.5	54.8	61.9	66.1	68.3	67.3	60.9	54.4	47.7	38.1		
	10%	DB	32.3	34.1	48.2	68.1	84.1	92.9	95.1	94.7	80.4	63.0	48.2	37.0		
		MCWB	32.0	32.9	42.2	54.0	61.4	65.6	67.7	66.6	60.1	52.1	45.0	35.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	40.7	53.1	62.4	69.7	73.7	78.5	76.8	69.2	63.2	55.6	43.6		
		MCDB	38.9	42.1	60.1	73.4	82.3	83.9	90.8	85.3	81.5	71.2	57.4	44.5		
	2%	WB	35.9	36.9	48.5	59.9	66.9	71.6	75.0	73.5	66.8	58.9	51.4	40.9		
		MCDB	36.1	37.8	56.4	71.9	80.6	84.5	85.4	86.5	78.1	65.5	53.6	41.7		
	5%	WB	34.0	34.9	45.4	57.8	65.3	70.0	72.9	71.0	64.5	56.1	48.0	38.0		
		MCDB	34.2	35.7	50.8	69.7	79.3	85.2	85.7	86.6	77.4	64.0	51.1	39.0		
	10%	WB	32.2	33.3	42.9	55.4	63.6	68.5	71.2	69.1	62.4	53.8	44.5	35.7		
		MCDB	32.3	34.2	46.7	64.7	77.7	84.5	86.4	87.6	75.8	61.8	47.5	36.3		
(35) Mean Daily Temperature Range	MDBR		12.1	14.7	16.2	20.7	22.7	24.4	24.3	25.3	23.2	20.1	14.1	10.0		
	5% DB	MCDBR	8.9	12.8	23.0	25.2	27.0	28.1	27.4	28.8	26.8	24.7	18.4	11.0		
		MCWBR	8.4	10.5	14.2	11.5	11.0	10.8	10.7	11.6	12.4	13.8	13.0	9.4		
	5% WB	MCDBR	8.8	11.4	19.9	22.2	22.8	23.6	23.6	24.7	23.7	20.9	15.8	9.5		
		MCWBR	8.3	10.0	13.4	11.2	11.0	11.7	11.8	13.1	13.5	13.7	12.4	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.358	0.376	0.434	0.496	0.480	0.454	0.455	0.460	0.417	0.398	0.372	0.369		
	taud		2.053	2.007	2.007	1.920	2.030	2.164	2.184	2.139	2.202	2.184	2.167	2.077		
	Ebn at Noon		213	239	244	241	251	259	257	249	249	231	209	191		
	Edh at Noon		31	40	47	56	52	46	44	45	38	33	28	27		
(44) All-Sky Solar Radiation	RadAvg		304	568	1039	1549	1965	2181	2100	1851	1407	854	443	258		
	RadStd		53	100	126	118	144	156	102	87	87	75	56	40		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)	N/A	N/A	N/A	+2.39	N/A	-1.31	N/A	N/A	+416	+279	

Nomenclature: See separate page